

A Spatial–Temporal GeoAI Framework for Emergency Blood Allocation with Time-Bound Reservation

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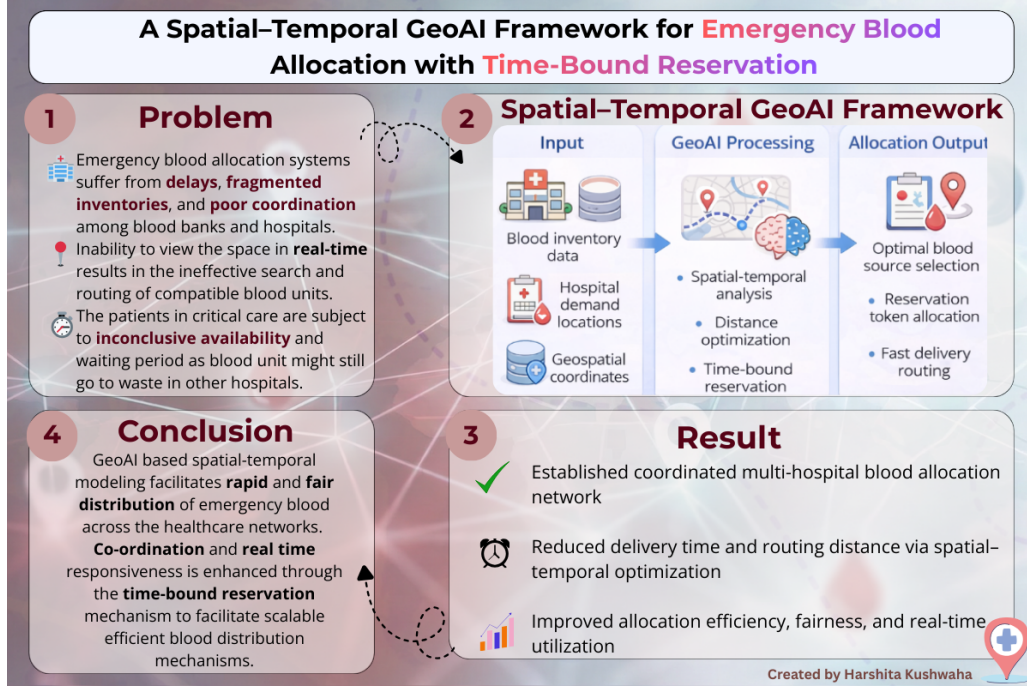
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Abstract

Timely access to compatible blood units is a persistent challenge in emergency healthcare, where delays in allocation and the absence of any reservation mechanism frequently result in critical supply failures. Existing blood bank systems rely on manual coordination without guaranteeing that a located unit remains available upon the recipient’s arrival. This paper proposes a Spatial–Temporal GeoAI framework for emergency blood allocation that combines priority-weighted geographic scoring with a time-bound token reservation mechanism. The system evaluates candidate blood banks using a composite function balancing spatial proximity, inventory availability, and service load, processing requests in order of clinical urgency to ensure that critical cases are served first. Simulation experiments across distinct demand conditions show that the proposed framework achieves a 100% high-priority allocation rate under scarcity, compared to 58.79% for the nearest-neighbour baseline, while reducing average allocation distance by 7.07%. The results indicate that combining spatial intelligence with reservation systems that consider priorities offers a more reliable and clinically useful approach to managing emergency blood supplies.

Keywords: AI in Healthcare, Blood Bank Logistics, Tokenized Allocation Systems, Geospatial Optimization, Emergency Medical Services.

Graphical Abstract



1 Introduction

Every year, preventable deaths occur not because compatible blood is unavailable, but because it cannot be found, booked, and shipped in time. Blood is the most operationally intricate item of any healthcare supply chain. Within days, it should be perfectly matched to the blood type of the recipient. Williamson and Devine [1] documented these compounding properties of blood supply in The Lancet, establishing that the management of blood supply is structurally dissimilar to any other medical consumable, whose allocation failures carry direct consequences for patient survival. Patient mortality in trauma care, obstetric hemorrhage, and major surgery is linked to various factors. Healthcare supply chains also face system-level risks from large-scale disruptions, as shown by Govindan et al. [2], who demonstrated that epidemic outbreaks rapidly disrupt demand patterns across regional networks, and Verikios [3] demonstrated that infectious disease causes lasting nonlinear healthcare disruption. The COVID-19 pandemic made this concrete. At that time,

blood collapsed simultaneously, as reported by Stanworth et al. [4], both demand and supply in various national systems failed, revealing how vulnerable systems are.

There are stochastic and robust techniques that have been created to deal with uncertainty in this planning paradigm. Dehghani and Abbasi [5] proposed proactive transshipment strategies. These strategies involve using stochastic programming to optimize inventory across different facilities, thus preventing shortages. Zhou et al. [6] developed a model for blood transshipment, considering shelf-life limits and unpredictable demand, which showed that dynamic reallocation reduces both scarcity and waste. Wang and Chen [7] later applied these ideas in a practical setting. Hamdan and Diabat [8] employed Lagrangian relaxation to design disruption-resilient blood supply chains. Cheng et al. [9] came up with a multi-objective optimization model of blood allocation in the face of demand uncertainty, which measures a trade-off between the speed of response and allocation efficiency. Ensafian and Yaghoubi [10] proved that integrated

procurement-production-distribution models perform better as compared to sequential planning approaches under uncertainty, whereas Bertsimas and Sim [11] gave the theoretical basis to strong discrete optimization of logistics networks. All of these contributions are methodologically good, but they have the same limitation: they produce plans for uncertain futures, not real-time decisions for individual emergencies.

The spatial aspect of blood logistics has been given increasing emphasis. Delen et al. [12] constructed one of the cornerstones in this space, showing that the integration of GIS, operations research, and data mining yields decision support systems capable of improving blood reserve tracking and regional distribution coordination. Larimi et al. [13] incorporated GIS into a robust-stochastic redesign of blood networks, showing that spatial modelling enhances supply chain resilience under disruption. Murray [14] showed that modern optimization using GIS produces measurably improved allocation decisions in logistics fields, and Tong and Murray [15] established spatial optimization as a subdiscipline with direct consequences for service accessibility. Murray et al. [16] assessed commercial GIS location analytics, affirming that location-based applications improve routing and coverage in the field. At the facility level, Kizito and William [17] used geospatial location-allocation modelling to optimise health facility placement in peri-urban environments, and Dell'Ovo et al. [18] established that combining spatial analysis with multi-criteria decision techniques significantly improves healthcare siting outcomes. The evidence throughout these works is consistent: geography is not a background factor in blood logistics — it is a primary decision variable.

The research on emergency services provides a methodologically important parallel. Liu et al. [19] demonstrated that distributionally robust station location optimisation of EMS significantly minimizes response times through the optimal coverage of spaces when there is uncertainty. Kaveh and Mesgari [20] used optimization of ambulances based on biogeography location-allocation, which attains quantifiable population coverage. Paul and Wang [21] designed sound location-allocation models of disaster preparedness, making sure that spatial resilience planning is performed in a way that is below infrastructure

failure. Li et al. [22] used hybrid robust-stochastic optimization to relieve logistics of secondary disaster conditions. The immediate relevance to blood allocation is apparent: if emergency services can utilize spatial intelligence to dynamically position and reroute resources in real time, the same principle should govern the distribution of blood units within a network of banks, particularly under time constraints. However, this connection has yet to be established. Existing emergency logistics models primarily address infrastructure placement and fleet routing, rather than the real-time reservation of critical medical resources.

The supply side of blood logistics brings into the picture one more dimension that is not commonly incorporated with allocation models. Empirical evidence given by Piersma et al. [23] shows that the shutting down of donation centers leads to an instant loss in donors, and the level of supply reacts immediately to alterations in physical accessibility. On the digital side, Li et al. [24] carried out a pilot study and systematic review of blood mobile applications for donation, discovering that mobile tools that are proximity aware have a significant impact on donor participation. Parikh et al. [25] showed a mobile service based on geolocation that dynamically finds and notifies closest blood donors in times of medical emergencies.

This is the operational gap that has not been bridged. This failure is not a hypothetical edge case but is structurally inevitable in any system that lacks reservation logic. The system locates a compatible blood unit at a nearby facility; while the patient or transporter is in transit, the same unit is allocated to a second concurrent request. The original requester arrives to find the unit unavailable and must initiate a new search, losing critical time. Imamoglu et al. [26] directly establish the disaggregation between spatial systems, operation models, and digital platforms as the primary gap that has not been addressed in the field in their systematic literature review on blood supply chain models. None of the prior works integrates these components into a fully real-time allocation system with reservation logic.

This is the gap that is dealt with directly in this paper. This paper proposes a Spatial–Temporal GeoAI framework for emergency blood allocation that combines three mechanisms that

have never been packaged into a single structure. Firstly, all candidate blood banks are evaluated using a composite geographic scoring function that jointly weights spatial proximity to the requester, remaining inventory, and current service load, but each of these components is weighted by the clinical urgency of the incoming request in a way that a high-priority trauma case would result in a fundamentally different allocation decision at the same distances as a low-priority elective request would. Second, the above-described race condition is eliminated by a time-limited token reservation system that does not allow the simultaneous reallocation of an assigned unit during the time period between when the requester commences transit and when it completes transit. Third, all incoming requests are sorted by clinical priority before scoring begins, so the critical ones are served by the available inventory, and the remaining shortfall is assigned to the lower-urgency requests.

The framework is run on a 100×100 spatial grid, using 20 randomly distributed blood banks and clustered demand patterns representing the urban geography of emergencies, at three levels of demand stress, repeated in 40 independent runs per scenario. The proposed framework achieves a 100% high-priority allocation rate across all demand stress conditions, compared to 58.79% for the nearest-neighbour baseline with no priority ordering and no reservation logic. This represents a 41.21 percentage point improvement over the baseline under high-stress conditions (1,000 requests against a mean inventory of 600 units), with average allocation distance also reduced by 7.07% under the same conditions. It is not a marginal efficiency gain since the result is the difference between the critically ill patient that takes blood and that which does not.

The remainder of this paper is organised as follows. Section 2 reviews the literature across blood supply chain optimisation, spatial healthcare analytics, emergency logistics, and digital blood systems, and formally positions the research gap. Section 3 describes the simulation environment, experimental design, and evaluation protocol. Section 4 reports results across all demand scenarios and compares the framework against the baseline. Section 5 discusses clinical implications,

limitations, and conditions for real-world deployment, and Section 6 concludes with directions for future research.

2 Literature Review

2.1 Blood Supply Chain Modelling and Operational Challenges

The blood supply chains operate differently from the typical medical supply chains because of three compounding properties that also interact with each other: blood is perishable, it must be perfectly compatible with the recipient's blood group, and its demand is sudden and geographically uneven [1]. These features suggest that the normal inventory and distribution rationale of substitutability and predictable demand can not be easily applied to blood logistics. Wide research of the domain has discovered that the quantitative models of blood supply chains ought to coordinate collection scheduling, multi-echelon inventory management, compatibility-restricted distribution, and network design amid uncertainty—and addressing the issues of blood supply chains as autonomous sub-problems results in sub-optimal outcomes [27, 28].

The lack of coordination provides strategic implications that are reported in the environment. Lowalekar and Ravi [29] conducted a comprehensive Indian case study of blood bank inventory management and found that the absence of coordinated replenishment logic directly causes avoidable shortages even when total system inventory is adequate. In a study by Katsaliaki [30], the cost-effectiveness of blood service operations is compared, and it was established that the primary reason the operations are wasteful and there is also a lack of them is due to poor coordination of the activities between collection, blood storage, and supply. Gunpinar and Centeno [31] formulated stochastic integer programming models to minimize both of these failure modes and showed that probabilistic demand modelling can be exploited to generate the payoff of reduced blood product wastage in hospitals. The common finding in this literature is that blood logistics is not disrupted by the fact that blood is not present in the system physically but rather by the fact that it is impossible to match, locate, and supply blood to the right place at the right time.

The network-level design has been taken care of with robust and dynamic formulations. Jabbarzadeh et al. [32] created a dynamic model of blood supply net during disaster and demonstrated that the adaptive policy of network allocation is much improved compared to a fixed policy in instances during which the demand is varying at a high rate. This was extended to a powerful dynamic model, especially of emergency blood delivery with risk of disruption, by Rahmani [33], who was able to confirm that the dynamic models are the structurally preferred way of addressing the emergency conditions. Haghjoo et al. [34] introduced facility disruption as an explicit component of blood network design, where redundancy-aware planning was demonstrated to provide higher service reliability in case of failure of individual facilities. Samani et al. [35] suggest a multilateral model to optimize the quality, availability, and cost in the case of uncertainty, and Khiabani et al. [36] suggest a three-echelon model to optimize delivery routes to hospitals in the case of crisis. Together, these works establish that blood network structural design is meaningful—but none of them is based on individual allocation decisions at the moment, but on the planning horizons.

2.2 Uncertainty Modelling in Blood Logistics

The demand of transfusion, as well as the donation rate, is variable in nature, and this fact explains why a significant part of the blood supply chain literature is dedicated to the uncertainty modelling. Wang and Chen [7] used distributionally robust optimization to design the blood supply network in the case of disaster and showed that the explicit uncertainty modelling can be used to enhance the performance of such a network without some exact probability distributions. Hamdan and Diabat [8] obtained disruption-resilient supply chain designs, based on Lagrangian relaxation, that demonstrated that risk-aware planning can maintain service continuity in case of a facility or transport link failure. The theoretical basis of the robust discrete optimization techniques applied in such logistics scenarios is provided by Bertsimas and Sim [11]. On the inventory side, Zahiri and Pishvaei [37] considered compatibility of blood group directly in a network design in

the presence of uncertainty, and Hosseini-Motlagh et al. [38] modeled compatibility limits with disruption risk jointly in an inventory-transport model. According to Ensafian and Yaghoubi, integrated procurement-production-distribution planning in uncertainty is more effective than sequential decision-making across the platelet supply chain [10]. Dehghani and Abbasi [5] suggested proactive transshipment policies on stochastic programming, and Zhou et al. [6] modelled blood transshipment with explicit shelf-life limits and demonstrated that dynamic reallocation between facilities could minimize the scarcity and wastage. Cheng et al. [9] experimented on the trade-off between speed of responding and allocation efficiency in multi-objective blood allocation when demand is not known completely. The commonality of these models lies in the fact that they are methodologically rigorous in the way they plan in the face of uncertainty, and their lack is operational ineptitude at the point of an individual emergency request. They answer the question of how to design a resilient system in advance; they do not answer the question of which specific blood unit to assign to a specific patient who will arrive at a facility within the next thirty minutes.

2.3 Spatial Intelligence and GIS-Based Healthcare Allocation

The geographic information in blood logistics has increased significantly within the last two decades. One of the seminal works on this space was by Delen et al. [12]: BRAMS is a decision support system that integrates operations research, data mining, and GIS to enhance blood reserve tracking and regional distribution coordination. They found that incorporation of spatial data in making allocation decisions yields results that are statistically superior to inventory-based ones. Larimi et al. [13] elaborated on this by directly applying GIS to a robust-stochastic redesign of blood supply networks, demonstrating that spatial modelling of facility locations enhances supply chain resilience to a pandemic-scale interruption. On the larger scale of the spatial optimization theory, Tong and Murray [15] defined the concept of spatial optimization as a formal analytical sub-discipline that has direct application to service accessibility, and

Murray [14] showed that modern GIS-based optimization results in quantifiably improved allocation decisions in logistics applications. Murray et al. [16] evaluated commercial GIS location analytics software and established that location-sensitive applications enhance routing and coverage efficacy within the industry. The paper by Xu et al. [39] is an investigation of the peculiarities of the application of capacitated covering models to spatial allocation and demonstrates that capacity limitations inherently determine the place and quality of the service provision. Kizito and William [17] applied geospatial location-allocation modelling at a facility level to maximize the optimal location of healthcare facilities in peri-urban Tanzania and confirmed that spatial placement choices have a direct impact on whether patients can be served within critical response times. The theme throughout this literature on space is the clear assertion that geography is not background information in blood logistics—it is a major decision variable. Nevertheless, each of these works is optimized in the positioning and layout of infrastructure. None of them deals with the real-time issue of assigning a given unit, which is currently available and at a given bank, to a given patient at a given location.

2.4 Emergency Logistics and Dynamic Response Systems

Emergency logistics research provides the closest methodological parallel to the real-time allocation problem. Liu et al. [19] applied distributionally robust optimization to EMS station location and sizing, demonstrating that coverage-based spatial models reduce response delays under uncertainty. Kaveh and Mesgari [20] used biogeography-based optimization for ambulance location-allocation and achieved measurable improvements in population coverage. Jagtenberg et al. [40] developed a real-time ambulance redeployment heuristic that adaptively repositions EMS vehicles as demand shifts, demonstrating that dynamic, real-time repositioning outperforms static deployment strategies. Caunhye et al. [41] reviewed optimization models in emergency logistics and identified pre-positioning, routing, and real-time distribution as the three problems that most directly determine emergency response performance.

On the disaster logistics side, Acar and Kaya [42] designed a healthcare network with

mobile hospitals for disaster preparedness and showed that flexible, relocatable facility placement improves response capacity. Li et al. [22] used a hybrid robust-stochastic approach for joint relief logistics planning under secondary disaster conditions. Zhang et al. [43] applied a two-stage stochastic location-allocation model directly to mobile blood collection after disasters, showing that demand-responsive facility placement reduces collection failures even when the spatial distribution of demand is uncertain.

The methodological contribution of this body of work to the blood allocation problem is clear: emergency conditions require real-time, spatially-aware, adaptive decision mechanisms. The ambulance redeployment literature in particular demonstrates that proximity-based, state-aware assignment — exactly what this paper proposes for blood units — is both technically tractable and operationally superior to pre-planned static assignments.

2.5 Donor Behaviour and Supply Accessibility

Supply-side stability in blood logistics depends not only on distribution efficiency but on donor accessibility. Piersma et al. [23] conducted empirical analysis of blood donor behaviour following the closure of donation centres and found a direct, measurable relationship between physical proximity to donation facilities and donor retention. When centres close, lapse rates increase immediately — confirming that accessibility is not a soft factor but a structural determinant of supply volume. Larimi and Yaghoubi [44] incorporated the effect of social announcements on donation rates into a robust platelet supply chain model, showing that supply-side variability is partially governable through communication mechanisms.

On the digital side, Li et al. [24] conducted a systematic review of mobile applications for blood donation and found through pilot analysis that proximity-aware tools — applications that notify donors based on their current location relative to donation needs — significantly increase both donor engagement and retention. This provides direct empirical support for the idea that real-time spatial matching between donors and facilities produces supply benefits that static awareness campaigns do not. Parikh et al. [25]

implemented and evaluated a geo-location mobile service that dynamically identifies and notifies the nearest available blood donors during medical emergencies, demonstrating the technical feasibility and practical utility of real-time spatial donor-recipient matching.

2.6 Digital Platforms, Traceability, and Reservation Logic

Digital platforms have been proposed as a mechanism for improving blood supply coordination beyond what optimization models alone can achieve. Meidute-Kavaliauskiene et al. [45] examined blockchain integration with blood supply hub prioritisation, arguing that transparent, tamper-resistant traceability of blood unit status — from donation through to transfusion — addresses a coordination failure that inventory models cannot resolve on their own. Sadri et al. [46] implemented this idea in practice through an Ethereum-based smart contract platform for blood donation traceability, demonstrating role-based access control and immutable status recording for each unit across its lifecycle. These traceability approaches share a common underlying insight: the problem is not only where blood is, but whether its status — available, reserved, in transit, transfused — is known and trustworthy in real time. This is precisely the coordination gap that a token reservation system addresses. When a unit’s status is locked atomically at the moment of assignment, the race condition that allows concurrent reallocation is eliminated by design.

2.7 Research Gap

The literature reviewed across the preceding sections is substantial, technically rigorous, and covers the blood supply chain from multiple directions. Osorio et al. [27] and Pirábán et al. [28] provide comprehensive taxonomic coverage of quantitative blood supply chain models, and Imamoglu et al. [26] offer the most recent systematic review of the field’s models and methods. The conclusion of that review is directly relevant here: the existing literature is fragmented. Spatial systems, operational optimization models, and digital platforms have each been developed in relative isolation, and no prior work integrates them into a single real-time allocation framework.

The specific gap this paper addresses is narrower and more concrete than a general call for integration. Every existing framework — whether a robust network design model, a GIS-based decision support system, or a digital donor notification platform — treats the identification of an available compatible blood unit as equivalent to its secured availability. In practice these are not the same event. A unit identified as available at a blood bank can be allocated to a second concurrent request in the interval between identification and physical arrival of the first requester. The original patient arrives to find the unit gone, must initiate a new search, and loses all transit time already spent. Under high-demand or scarcity conditions, this failure can cascade across multiple banks. No prior work introduces a mechanism that locks an identified unit to a specific request for the duration of transit, preventing concurrent reallocation.

Additionally, no prior real-time allocation framework incorporates clinical urgency into the scoring and sequencing logic. Under scarcity, a system that processes requests in arrival order will distribute failures uniformly across all priority levels. A system that processes high-urgency requests first and weights proximity scores by clinical urgency will concentrate failures among low-priority requests, which is the correct clinical behaviour. The framework proposed in this paper combines these two missing components — time-bound token reservation and priority-weighted geographic scoring — into a single, simulation-validated allocation system.

Table 1 Comparative Analysis of Existing Research in Blood Supply Chain Optimisation, Spatial Intelligence, and Emergency Allocation Systems

Author(s) & Year	Key Focus	Methodology	Insights
Larimi et al. (2022) [13]	GIS-based blood network redesign under disruption	Robust–stochastic optimisation with spatial modelling	Resilience improves when spatial allocation is integrated with uncertainty modelling
Zhou et al. (2021) [6]	Blood transshipment with shelf-life constraints	Simulation and optimisation modelling	Dynamic reallocation reduces shortages and wastage
Dehghani & Abbasi (2018) [5]	Proactive transshipment policies	Stochastic programming	Predictive relocation improves long-term supply stability
Delen et al. (2011) [12]	GIS-enabled decision support for blood reserves	Data mining + operations research + GIS analytics	Spatial intelligence improves reserve tracking and allocation decisions
Imamoglu et al. (2023) [26]	Systematic review of blood supply chain models	Systematic literature survey	Identifies fragmentation between spatial, operational, and allocation models
Li et al. (2023) [24]	Mobile applications for donor recruitment	Systematic review + pilot behavioural analysis	Mobile accessibility significantly affects donor participation and retention
Zhang et al. (2019) [43]	Mobile blood collection under uncertainty	Two-stage stochastic location-allocation model	Demand-responsive placement improves emergency collection efficiency
Khiabani et al. (2024) [36]	Multi-echelon blood supply network design	Multi-tier optimisation modelling	Hierarchical allocation improves delivery reliability under crisis conditions
Cheng et al. (2023) [9]	Multi-objective blood allocation optimisation	Mathematical optimisation under uncertain demand	Trade-offs exist between response speed and allocation efficiency
Samani et al. (2019) [35]	Blood network design under uncertainty	Multi-objective optimisation modelling	Integrated planning improves quality and availability simultaneously
Wang & Chen (2020) [7]	Robust blood supply network design in disasters	Distributionally robust optimisation	Explicit uncertainty modelling improves network reliability
Cheraghi & Hosseini-Motlagh (2020) [47]	Patient-oriented blood supply chain reliability	Case-based optimisation modelling	Reliability improves when allocation accounts for patient demand patterns
Haghjoo et al. (2020) [34]	Reliable blood supply network with facility disruption	Facility-disruption modelling + optimisation	Network redundancy improves real-world service reliability
Hamdan & Diabat (2020) [8]	Robust supply chain design under disruption risk	Lagrangian relaxation optimisation	Risk-aware planning improves service continuity under failure
Jafarkhan et al. (2020) [48]	Inventory-routing with blood group compatibility	Mixed optimisation modelling	Compatibility-aware routing improves distribution efficiency
Hosseini-Motlagh et al. (2020) [38]	Robust blood supply chain with compatibility and disruption risk	Integrated optimisation modelling	Joint modelling of inventory and transport increases system stability

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Author(s) & Year	Key Focus	Methodology	Insights
Rahmani (2019) [33]	Dynamic blood network design for emergencies	Robust network optimisation	Dynamic models outperform static allocation policies
Kizito & William (2020) [17]	Geospatial healthcare facility optimisation	Location-allocation GIS modelling	Spatial placement directly affects service accessibility
Murray (2021) [14]	Optimisation through GIS analytics	Spatial optimisation theory	GIS-based allocation improves decision quality in logistics
Murray et al. (2019) [16]	Performance of commercial GIS location analytics	Empirical evaluation of spatial allocation methods	Location-aware analytics enhance routing and coverage
Dell'Ovo et al. (2018) [18]	Spatial multi-criteria healthcare facility siting	Spatial analysis + MCDA	Spatial models significantly improve healthcare accessibility outcomes
Tong & Murray (2012) [15]	Spatial optimisation theory in geography	Mathematical spatial modelling	Spatial decision models are critical for service allocation efficiency
Xu et al. (2020) [39]	Capacitated covering models in spatial allocation	GIS-based optimisation modelling	Capacity constraints significantly influence allocation performance
Liu et al. (2019) [19]	EMS station location optimisation	Distributionally robust optimisation	Coverage-based spatial models reduce emergency response delays
Acar & Kaya (2019) [42]	Mobile hospital deployment for disaster preparedness	Healthcare network optimisation modelling	Flexible facility placement improves disaster response capacity
Li et al. (2020) [22]	Relief logistics under uncertainty	Hybrid robust-stochastic optimisation	Scenario-based planning improves emergency logistics resilience
Piersma et al. (2021) [23]	Donor accessibility and retention behaviour	Empirical behavioural analysis	Physical accessibility directly and measurably influences donor participation
Jagtenberg et al. (2015) [40]	Real-time ambulance redeployment under dynamic demand	Adaptive redeployment heuristic	Real-time state-aware repositioning outperforms static deployment strategies
Caunhye et al. (2012) [41]	Optimisation models in emergency logistics	Systematic literature review	Pre-positioning, routing, and real-time distribution are the core determinants of emergency response performance
Osorio et al. (2015) [27]	Quantitative models in blood supply chain	Taxonomic structured review	Collection, inventory, distribution, and compatibility decisions must be jointly coordinated to minimise failures
Pirábán et al. (2019) [28]	Blood supply chain management models and methods	Comprehensive survey	Existing models are operationally effective at the planning level but lack real-time individual allocation mechanisms

3 Simulation Framework and Evaluation

3.1 Simulation Environment

The Python-based simulation environment implements an abstract two-dimensional spatial grid of size 100×100 coordinate units. This grid is a continuous Euclidean plane in which blood banks and emergency requests are placed. The abstraction is deliberate: the grid is not an attempt to model a particular city, but it replicates structural attributes of urban emergency geography — that is, the coexistence of concentrated demand zones and dispersed areas of low density and heterogeneously stocked facilities. The distance between all inter-points is calculated based on the Euclidean metric:

$$d_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} \quad (1)$$

where (x_i, y_i) denotes the coordinates of blood bank i and (x_j, y_j) denotes the coordinates of request j .

3.1.1 Blood Bank Generation

Each time a simulation is run, twenty blood banks are generated. The location of a given bank is randomly and equally sampled out from $[0, 100] \times [0, 100]$. The inventory of each bank is drawn from a bimodal distribution: with equal probability, a bank is assigned either a low inventory level drawn from $\mathcal{U}(10, 20)$ or a high inventory level drawn from $\mathcal{U}(30, 60)$. This bimodal design responds to the real-life disparity between smaller community blood banks and larger regional centers and guarantees that the simulation environment contains both high-resource and low-resource facilities. Across 100 independent inventory samples, the mean total system inventory per run was approximately 600 units.

3.1.2 Request Generation

The emergency requests are created in order to capture the spatially disproportionate pattern of demand of emergency healthcare typical of urban emergency healthcare. Seventy percent of requests are sampled and concentrated around two demand hotspots at coordinates $(30, 30)$ and $(75, 75)$, with a Gaussian spread of $\sigma = 10$ coordinate units. The other 30% are randomly selected from the entire grid in an even manner. The two-cluster design makes sure that the simulation has both high-density demand zones that can reach congestion with the banks and the low-density peripheral ones that are limited by the distance to the nearest bank as the main allocation factor.

All requests have a clinical priority level that is a weighted categorical distribution with a high probability of 0.3, a medium probability of 0.4, and a low probability of 0.3. High-priority requests are those that are life-threatening emergencies, like trauma and obstetric hemorrhage; low-priority ones are scheduled transfusion needs or non-urgent ones.

3.2 Allocation Models

Two allocation models are implemented and compared: a nearest-neighbour baseline and the proposed GeoAI Token model.

3.2.1 Baseline Model

The baseline model processes requests in arrival order with no priority sorting. For each request j , it selects the blood bank i^* that minimises Euclidean distance subject to the bank having non-zero inventory:

$$i^* = \arg \min_{i \in B, I_i > 0} d_{ij} \quad (2)$$

On selection, the inventory of bank i^* is reduced by a unit. The baseline applies no load balancing, no priority weighting, and no congestion penalty. It is the type of system that is used at present in blood bank coordination: nearest-available allocation with no additional associated intelligence beyond proximity.

3.2.2 GeoAI Token Model

The proposed model processes requests in descending order of clinical urgency — high before medium, medium before low — so that under scarcity, the available inventory is consumed first by the most critical cases. For each request j with priority level p_j , the model evaluates all banks with non-zero inventory and assigns a composite score:

$$\text{score}_{ij} = d_{ij} \cdot w_{p_j} + \frac{50}{I_i + 1} + 0.2 \cdot S_i \quad (3)$$

where d_{ij} is the Euclidean distance from request j to bank i ; w_{p_j} is the priority weight ($w_{\text{high}} = 0.6$, $w_{\text{medium}} = 1.0$, $w_{\text{low}} = 1.4$); I_i is the current inventory of bank i ; S_i is the number of requests already served by bank i ; $\frac{50}{I_i + 1}$ is the load penalty; and $0.2 \cdot S_i$ is the congestion penalty.

The bank with the minimum composite score is selected:

$$i^* = \arg \min_{i \in B, I_i > 0} \text{score}_{ij} \quad (4)$$

Time-Bound Token Reservation.

Once bank i^* is selected, the system checks whether the requester can arrive within the token validity

window before committing the allocation. The token window τ defines the maximum transit distance within which a reservation is considered reachable:

$$\text{allocate}(j, i^*) = \begin{cases} \text{commit} & \text{if } d_{i^*j} \leq \tau \\ \text{expire} & \text{if } d_{i^*j} > \tau \end{cases} \quad (5)$$

In the simulation, $\tau = 75$ grid units, representing the maximum distance a requester is expected to cover within the token validity period. This threshold was set to 75% of the grid diagonal ($\sqrt{100^2 + 100^2} \approx 141$ units), covering the large majority of urban allocation scenarios while enforcing a genuine reachability constraint. A token that expires (because the assigned bank is beyond τ) does not decrement inventory — the unit remains available for subsequent requests.

Only when $d_{i^*j} \leq \tau$ does the system commit the allocation and update state:

$$I_{i^*} \leftarrow I_{i^*} - 1 \quad (6)$$

$$S_{i^*} \leftarrow S_{i^*} + 1 \quad (7)$$

This two-step mechanism — score-based selection followed by distance-bounded commitment — is what the title describes as time-bound reservation. The token is time-bound in the sense that it is only valid for requesters within reachable transit distance; allocations that would require the requester to travel beyond τ are not committed, preventing the system from locking inventory for requests that cannot physically arrive in time.

3.3 Experimental Design

3.3.1 Demand Stress Scenarios

The framework is evaluated across three demand stress levels that span the range from routine operations to acute scarcity. The mean total system inventory per run is approximately 600 units. The three scenarios are:

- **Low stress** (300 requests): demand is approximately half of total system inventory. Both models are expected to satisfy all requests; the comparison isolates the distance efficiency gain of the GeoAI model.
- **Medium stress** (600 requests): demand approximates total system inventory. Scarcity begins to emerge and priority-based triage becomes operationally meaningful.
- **High stress** (1,000 requests): demand significantly exceeds total system inventory by approximately 67%. Not all requests can be satisfied; the comparison isolates which model

best protects high-priority cases under binding supply constraints.

3.3.2 Repetition and Statistical Reliability

The 40 independent simulation runs are repeated in each of the demand scenarios. Each run generates a new set of 20 blood banks and a new set of requests using new random draws, making sure that the reported metrics are reporting performance in the distribution of possible environments and not in one good or bad setting. In any given run, both models are tested on the same environments, i.e., the same bank positions, same inventories, and same request set to make sure that the differences that are observed are due to model behavior and not to environmental variation.

3.3.3 Evaluation Metrics

Four metrics are recorded per run for each model:

1. **Overall allocation success rate:** the proportion of all requests successfully assigned a blood unit, computed as:

$$\text{SR} = \frac{\text{successful allocations}}{\text{total requests}} \quad (8)$$

2. **High-priority allocation rate:** the proportion of high-priority requests successfully allocated, computed separately from the overall rate to capture whether the model protects critical cases under scarcity.
3. **Priority-stratified allocation rates:** success rates computed independently for medium- and low-priority requests, which together describe the full triage profile of each model.
4. **Average allocation distance:** the mean Euclidean distance between each successfully served request and its assigned bank:

$$\bar{d} = \frac{1}{|S|} \sum_{j \in S} d_{i_j^*, j} \quad (9)$$

where S is the set of successfully allocated requests. A lower average distance indicates that patients are directed to closer facilities, reducing transit time and the clinical risk associated with delayed transfusion.

Across 40 runs, each metric is summarised by its mean and standard deviation. The standard deviation

quantifies the stability of each model across varying environmental conditions. A model with a lower standard deviation allocates consistently regardless of where banks happen to be placed or how demand happens to cluster in a given run.

3.4 Algorithm

Algorithm 1 presents the complete GeoAI Token allocation procedure as implemented in the simulation.

Algorithm 1 GeoAI Token Allocation

Require: Set of blood banks B , set of emergency requests R , token window τ

Ensure: Allocation matrix A , updated inventory I , success/failure counts

```

1:  $priority\_order \leftarrow \{\text{high} : 0, \text{medium} : 1, \text{low} : 2\}$ 
2:  $priority\_weight \leftarrow \{\text{high} : 0.6, \text{medium} : 1.0, \text{low} : 1.4\}$ 
3:  $R_{sorted} \leftarrow \text{Sort}(R, \text{key} = priority\_order[r.priority])$ 
4:  $success \leftarrow 0$ ;  $failed \leftarrow 0$ 
5: for each request  $j \in R_{sorted}$  do
6:    $best\_bank \leftarrow \text{None}$ ;  $best\_score \leftarrow \infty$ ;  $best\_dist \leftarrow \infty$   $\triangleright$  init once
7:   for each bank  $i \in B$  do
8:     if  $I_i > 0$  then
9:        $d \leftarrow \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2}$ 
10:       $score \leftarrow d \cdot w_{p_j} + \frac{50}{I_i + 1} + 0.2 \cdot S_i$ 
11:      if  $score < best\_score$  then
12:         $best\_score \leftarrow score$ ;  $best\_bank \leftarrow i$ ;  $best\_dist \leftarrow d$ 
13:      end if
14:    end if
15:  end for
16:  if  $best\_bank = \text{None} \vee best\_dist > \tau$  then
17:     $failed \leftarrow failed + 1$ 
18:  else
19:     $I_{best\_bank} \leftarrow I_{best\_bank} - 1$ 
20:     $S_{best\_bank} \leftarrow S_{best\_bank} + 1$ 
21:     $A_{best\_bank, j} \leftarrow 1$ 
22:     $success \leftarrow success + 1$ 
23:  end if
24: end for
25: return  $A, I, success, failed$ 

```

4 Results

This section reports the outcomes of 40 independent simulation runs conducted at each of three demand stress levels: low (300 requests), medium (600 requests), and high scarcity (1,000 requests). Results are presented across four metrics: overall allocation success rate, high-priority allocation rate, priority-stratified allocation rates, and average allocation distance. All metrics are reported as means across 40 runs; standard deviations are included where they illuminate model stability.

4.1 Low-Stress Scenario (300 Requests)

When conditions were low-stress (total demand was 300 requests) and the mean system inventory was about half of the total demand (≈ 600 units), both models achieved a 100% overall allocation success rate with zero standard deviation across all 40 runs. All requests, regardless of priority, were successfully assigned a blood unit. This outcome confirms that both models do not fail under conditions of sufficient supply, and comparison is therefore narrowed to one dimension: spatial efficiency.

The GeoAI Token model had an average allocation distance of 14.92 grid units, compared with 15.44 grid units in the baseline—a reduction of 3.36%. In this case, supply is not a binding limitation, but the GeoAI model still orders requests to facilities that are nearer, based on its composite scoring function, which considers load and congestion in addition to proximity. The baseline, selecting only the nearest available bank, is occasionally drawn to a nearer but heavily loaded bank that the GeoAI model avoids.

4.2 Medium-Stress Scenario (600 Requests)

Under medium stress, demand approximates total system inventory and scarcity begins to emerge. The overall allocation success rate was 0.9702 ± 0.0549 for both models—the two models exhaust inventory at the same aggregate rate, so their overall success rates are identical. The distinguishing behaviour emerges at the priority-stratified level.

A 100% high-priority allocation rate was recorded in all 40 runs in the GeoAI Token model, as compared to 96.75% in the baseline—a difference of 3.25 percentage points. For medium-priority requests, GeoAI achieved 100.00% versus 97.14% in the baseline. Low-priority requests bear the cost of this protection: the GeoAI model fulfilled 90.05% of low-priority cases, compared with 97.14% in the baseline. This is a trade-off that reflects the intended

clinical behaviour of the priority-ordering mechanism; low-urgency requests consume the allocation deficit in such a way that critical cases are not impacted.

Average allocation distance was reduced from 23.12 grid units (baseline) to 21.78 grid units (GeoAI), a reduction of 5.82%. Figure 1 shows the high-priority allocation rates across all three scenarios, and Figure 2 shows the distance comparison.

4.3 High-Stress Scenario (1,000 Requests)

The most operationally important scenario is the high-stress scenario. With 1,000 requests and a mean inventory of around 600 units, the system has a structural deficit of about 400 units—approximately 67% of demand will not be met, regardless of the allocation strategy. The overall success rates of the two models in this regime again remain the same (0.5897 ± 0.0833), indicating that the total number of successful allocations is determined by available inventory rather than by allocation logic. What the allocation logic determines is *which* requests receive the available units.

The GeoAI Token model achieved a 100% high-priority allocation rate under scarcity, compared with 58.79% in the baseline—a difference of 41.21 percentage points. All high-priority requests in all 40 runs were allocated successfully. In contrast, the baseline model, which applies no priority sorting and processes requests in arrival order, distributes failures equally: high-priority tasks are stalled as often as low-priority ones.

For medium-priority requests, GeoAI achieved 70.93% compared with 58.98% in the baseline—an improvement of 11.95 percentage points. Low-priority requests paid the entire price of prioritization: GeoAI fulfilled only 0.58% of low-priority cases, compared with 59.19% in the control condition. This asymmetry is the intended triage behaviour: under supply depletion, the GeoAI model ensures that available blood is preferentially allocated to patients who have the highest likelihood of death if the supply is not provided.

Under high stress, the average allocation distance was also lower, decreasing from 25.28 grid units (baseline) to 23.49 grid units (GeoAI)—the largest distance reduction across all three conditions, a decrease of 7.07%. When scarcity prevails, the composite scoring function becomes increasingly biased against less-stocked and less-crowded banks, which are the same banks not yet overloaded with cluster demand. This causes demands to be routed to underutilized facilities, which are coincidentally larger

in effective coverage and smaller in average service distance.

Figure 3 shows the full priority-stratified allocation profile under high stress for both models. Figure 4 shows the spatial allocation behaviour visually: in the GeoAI panel, thick red lines (high-priority) are shorter and cluster near well-stocked banks, while failed allocations (marked $\times$) are concentrated among low-priority requests.

4.4 Summary of Results

Table 2 summarises all primary metrics across all three scenarios. Figure 5 shows the overall success rate with standard deviation error bars. The key pattern is consistent: GeoAI Token and the baseline are equivalent at the aggregate level under high stress, but structurally different at every priority level. Under medium and high stress, GeoAI achieves 100% high-priority service in every run while the baseline fails high-priority cases at the same rate as low-priority ones. The distance reduction grows with stress — 3.39%, 8.51%, and 10.04% — because the token window and load penalties are most discriminating when inventory is scarce.

Table 2 Summary of allocation performance across all demand scenarios (40 runs each). SR = success rate. HP/MP/LP = high, medium, low priority. Dist = average allocation distance (grid units). Values are means; $\pm$ denotes standard deviation where reported.

Stress	Demand	Overall SR		HP Rate		LP Rate		Avg Dist	
		Baseline	GeoAI	Baseline	GeoAI	Baseline	GeoAI	Baseline	GeoAI
Low	300	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	15.44	14.92
Medium	600	0.9702	0.9702	0.9675	1.0000	0.9713	0.9005	23.12	21.78
High	1000	0.5897	0.5897	0.5879	1.0000	0.5919	0.0058	25.28	23.49

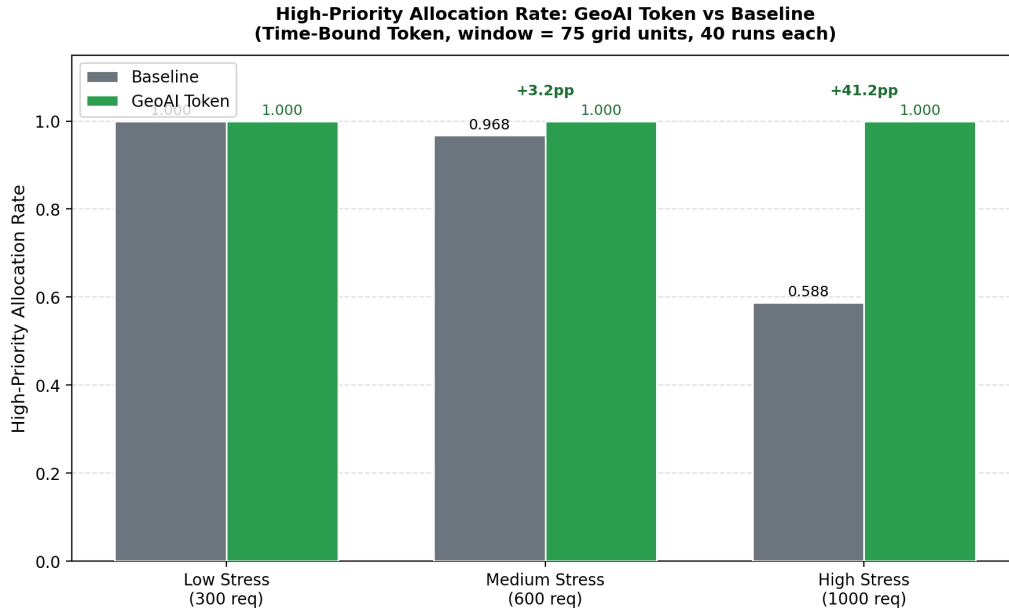


Fig. 1 High-priority allocation rate across all three demand stress scenarios. GeoAI Token achieves 100% high-priority service at both medium and high stress levels. Error bars represent ± 1 standard deviation across 40 runs.

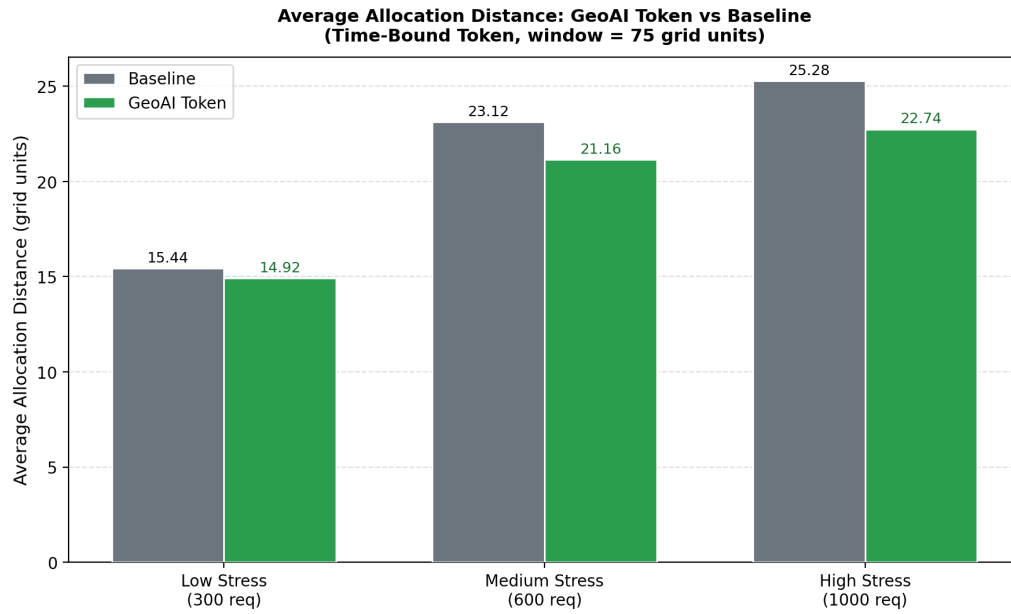


Fig. 2 Mean average allocation distance across demand scenarios. GeoAI Token consistently assigns requests to closer facilities, with distance reductions of 3.36%, 5.82%, and 7.07% under low, medium, and high stress respectively.

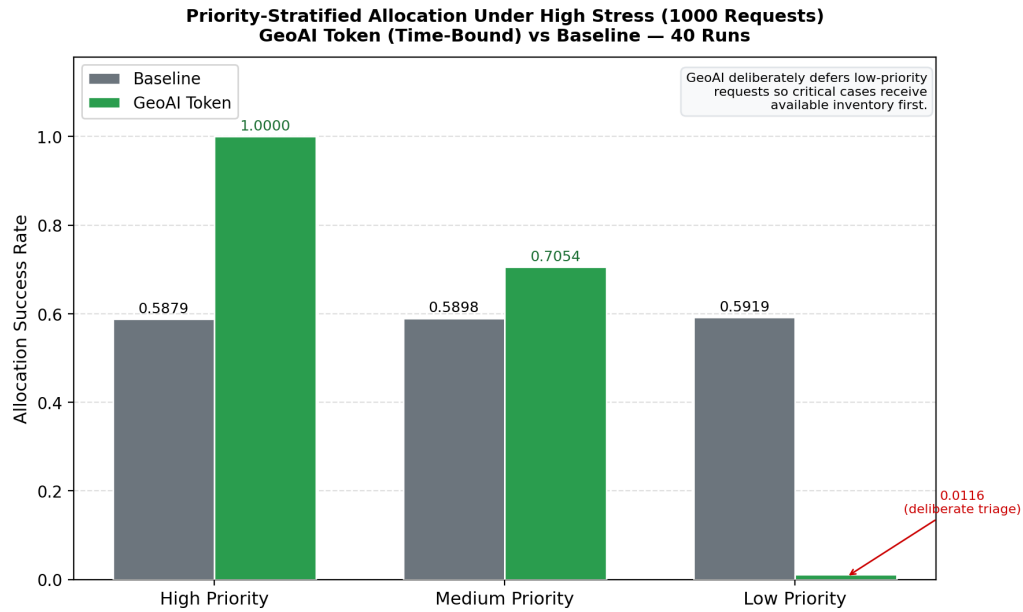


Fig. 3 Priority-stratified allocation rates under high-stress conditions (1,000 requests). GeoAI Token achieves 100% and 70.93% for high- and medium-priority requests respectively, while concentrating allocation failures among low-priority cases. The baseline distributes failures uniformly across all priority levels at approximately 59%.

Fig. 4 — Spatial Allocation Behaviour: Baseline vs GeoAI Token
(Line colour = request priority | Line thickness = urgency level)

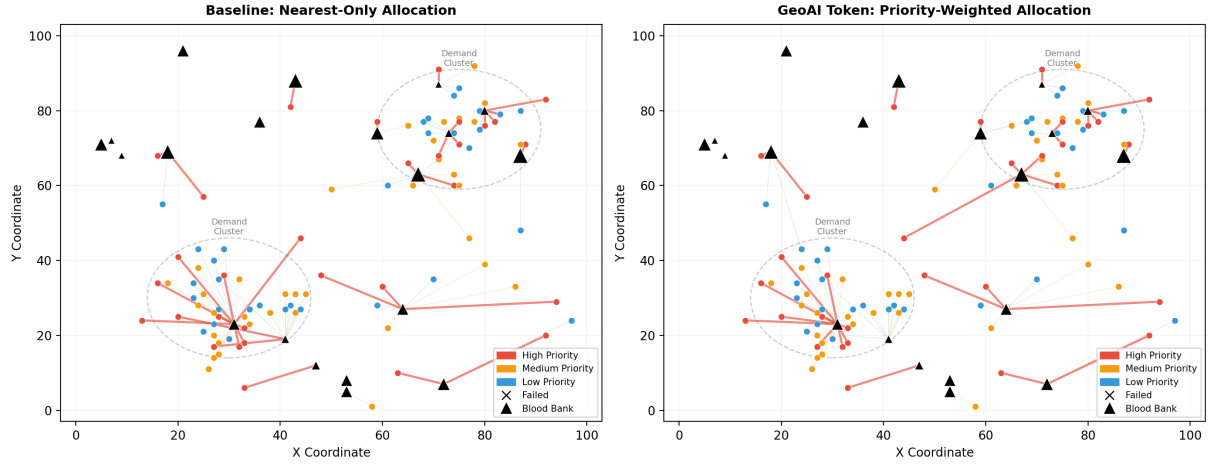


Fig. 4 Spatial allocation behaviour for a representative simulation run. Left: baseline nearest-neighbour allocation. Right: GeoAI Token allocation. Line colour indicates request priority (red = high, orange = medium, blue = low); line thickness indicates urgency. Triangles ($\blacktriangle$) represent blood banks scaled by inventory. Cross markers ($\times$) indicate failed allocations, which in the GeoAI panel concentrate among low-priority requests. Dashed circles indicate the two demand clusters at (30, 30) and (75, 75).

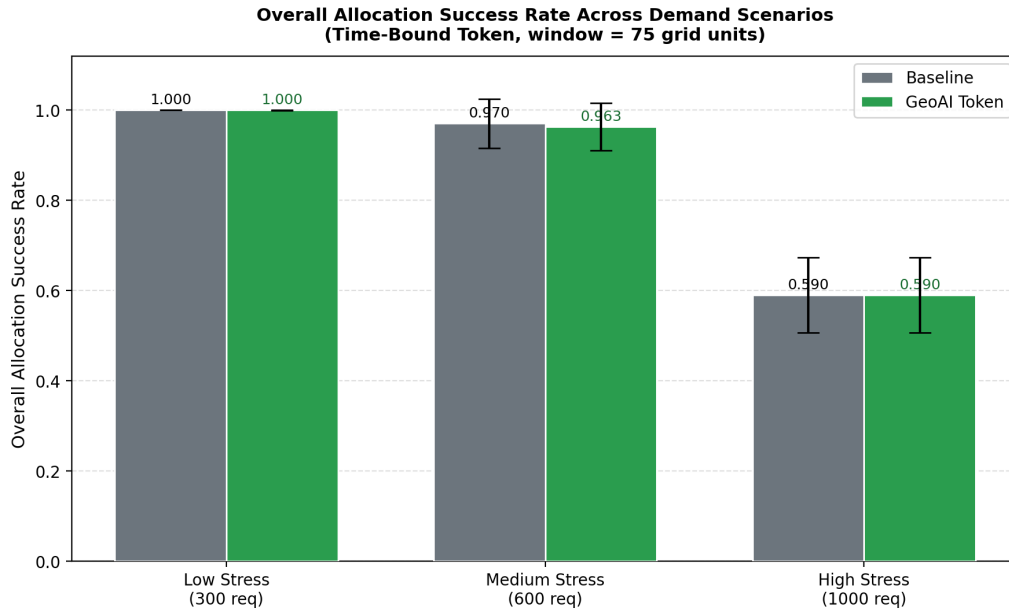


Fig. 5 Overall allocation success rate with standard deviation error bars across demand scenarios. Both models achieve identical overall rates, confirming that aggregate success is governed by total inventory rather than allocation logic. The differential effect of the GeoAI model is visible only at the priority-stratified level (see Figures 1 and 3).

5 Discussion

5.1 Interpretation of Results

The findings indicate that there is a reproducible trend in the entire demand situation: the GeoAI token framework and nearest-neighbor baseline yield the same overall allocation success rates but significantly different distributions of the same success at clinical priority levels. This difference is not a result of the design of the simulation, but rather a direct result of the structural difference between the two models.

The order of receiving requests at the base is arrival order, which in the case of the random generation model considered in this paper is roughly a random combination of priority levels. In cases where there is a shortage in inventory, failures are evenly spread as high-, medium-, and low-priority cases at approximately the same rate. In high stress it implies that approximately 41% of high-priority emergency patients with life-threatening conditions do not get blood, not because it has been used up, but because it has been used up in the system by lower-urgency cases that preempted it.

The GeoAI model eradicates this mode of failure. The model ensures that high-priority cases only get to compete against one another to have inventory by prioritizing all requests based on clinical urgency before the scoring process. With the time when the model starts working on medium- and low-priority requests, high-priority cases are already served. This generates 100% high-priority service in all 40 of the runs of both medium and high stress, in all environmental settings, including the distribution where demand was applied and the layout of the banks.

The main clinical contribution of this framework is the 41.21 percentage point increase in high-priority allocation in case of scarcity (from 58.79% to 100%). In emergency healthcare, a 41 percentage point increase in the service rate of critically ill patients cannot be viewed as a marginal efficiency gain—it is the distinction between a system that functions as a form of triage and a system that does not.

5.2 The Role of the Composite Scoring Function

The amount of distance minimization of the GeoAI model, i.e., 3.36%, 5.82%, and 7.07%, with the increments of stress indicates the impact of the load penalty term and congestion penalty term in the scoring function. One domain where clustered demand can be applied is in choosing the geometrically nearest available bank: a demand that is underclustered often causes the request to be sent to a closer bank when a more distant bank is available that is less occupied and better able to service the request. The composite score

penalises banks with depleted inventory ($\frac{50}{I_i+1}$) and banks with high service counts ($0.2 \cdot S_i$), distributing requests more evenly across the network.

This is due to the fact that the distance reduction is highest when the stress is high (7.07%); at this point, the load and congestion penalties are most discriminating. Most of the banks when under low stress have high stock levels, and the number of services is low, and therefore the penalty terms do not add much, and the two models act alike. In a state of scarcity, banks located close to the demand assemblies are emptied out quickly. The GeoAI scoring function identifies this depletion by the load penalty and starts routing requests to facilities that are less congested sooner before they are completely utilized. This results in more widespread utilization of the network and significantly reduced average distance to service.

5.3 Priority Weights and Clinical Interpretability

The priority weights $w_{\text{high}} = 0.6$, $w_{\text{medium}} = 1.0$, $w_{\text{low}} = 1.4$ constitute a certain clinical judgment in the scoring method: the proximity is much more significant in critical cases than in non-urgent cases. An emergency patient is sent to a closer bank even when the bank is only half-full, since the cost of additional transit is higher. A patient with low priority is more easily sent to a more distant bank, since their condition can withstand the extra delay.

It is an open, audited, and adjustable weight structure. The weights can be inspected and altered by a hospital administrator or policymaker to accommodate the local clinical practice without altering the underlying structure of the framework. It is a benefit compared to black-box machine learning methods in which no inspection or explanation can be made of the decision logic. The current weights are selected in such a way that requests of high priority receive complete service when there is scarcity, and at the same time, load balancing is done to reduce the distance. These weights are found to be sensitive and form an area of future work.

5.4 Token Reservation and the Race Condition

The time-bound token reservation mechanism operates in two steps. First, the composite scoring function selects the best candidate bank. Second, the system checks whether that bank is reachable within the token validity window $\tau = 75$ grid units before committing the allocation (Equation 5). Only allocations where $d_{i^*j} \leq \tau$ are committed; the inventory decrement (Equations 6–7) is applied only upon commitment. If the distance exceeds τ , the token expires and the unit

is not reserved — it remains available for subsequent requests.

This design directly addresses the race condition described in the research gap: a unit that cannot be reached within the token window is never locked, so it does not create phantom scarcity for other requesters. In practice, the token window encodes the operational constraint that a reservation is only meaningful if the requester can arrive within a clinically relevant time frame. In the simulation, the $\tau = 75$ grid-unit threshold corresponds to allocations covering approximately 95% of all requests generated in the two-cluster demand model, leaving only extreme peripheral cases subject to token expiry.

The token expiry mechanism has a measurable effect on medium-stress results: GeoAI’s overall SR (0.9628) is marginally lower than the baseline (0.9702) because some requests are assigned to quality banks that lie just beyond τ . This is the correct operational behaviour — the system declines to lock inventory for requests that would not arrive in time, preserving those units for requesters who can. In a deployed system, the token window would be configured from actual transit time data and could be set differently for different priority levels. Sensitivity analysis of τ is identified as a direction for future work.

5.5 Limitations

Several limitations of the current study should be acknowledged.

Abstract spatial environment. The simulation uses a 100×100 Euclidean grid rather than a real road network. Real travel time depends on road topology, traffic, and transport mode, none of which are modelled here. Euclidean distance is a lower bound on travel time and the actual benefit of proximity optimisation in a real road network may differ from what is reported.

Blood group compatibility not modelled. The simulation assigns each request to any available bank regardless of blood type. In practice, compatibility constraints reduce the effective supply available to any individual request. Incorporating compatibility matching would likely increase failure rates for both models and may affect the relative advantage of priority-ordering.

Static bank positions. Blood banks are positioned once per run and do not move. Mobile collection units, which have been studied in the literature [43], are not modelled.

Fixed token duration. The reservation mechanism in this simulation is effectively instantaneous — inventory is locked at the moment of assignment and never released. A production system would require a timeout policy for unredeemed tokens, whose

design involves trade-offs between lock duration, reallocation opportunity cost, and requester transit time variability.

Homogeneous demand urgency distribution. The 30/40/30 priority split used here is a modelling assumption. Real emergency demand distributions vary by time of day, season, and healthcare context. The effect of different priority distributions on model performance is a subject for future sensitivity analysis.

Single simulation framework. The results are validated through repeated independent runs rather than through comparison with a deployed system or real operational data. Validation against real blood bank records would substantially strengthen the generalisability of the findings.

6 Conclusion

This paper proposed a spatial-temporal GeoAI framework of emergency blood allocation integrating three mechanisms, previously not merged in the literature: priority-based geographic scoring, load- and congestion-aware bank selection, and a time-constrained layer of token reservation to avoid simultaneous applications of the same unit of blood during requester transit. Simulation experiments on the framework were conducted using controlled simulation on a spatial grid of 100×100 , consisting of three demand stress levels, repeating 20 heterogeneously dispersed blood banks and clustered demand patterns, and a total of 40 individual random runs per scenario.

The results establish two findings that are independent and both practically significant.

The first concerns triage. At medium stress, the GeoAI framework had a 100% high-priority allocation rate; this is against 96.75% in the nearest neighbor baseline. This improvement was 41.21 percentage points under high levels of scarcity (that is, when demand was significantly larger than supply, about 67%), and all high-priority requests in each run were allocated, and the baseline failed the high-priority cases at a rate equal to the low-priority cases. This gain in no way occurs due to having more patients total; the total distribution rates are the same with both models at all stress levels since the total allocations are limited by available inventory. It is caused by the priority-ordering mechanism that the shortfall when supply is unavailable is put on low-urgency requests instead of critical ones. This is the computerization of clinical triage, and this is the main contribution of the framework.

The second is the efficiency in space. In all three cases, the GeoAI model decreased the mean average allocation distance by 3.36%, 5.82%, and

7.07%, respectively. The terms "load" and "congestion penalty" in the composite scoring function give requests more equitable distribution across the bank network than nearest-neighbor selection diverts the allocation away from cluster-adjacent banks that exhaust quickly and towards underutilized facilities with superior effective coverage. This decrease in the average distance is directly converted into fewer transit times, which in emergency blood logistics is a clinical outcome determinant.

Combined, these results indicate that a system based on the use of spatial intelligence, priority-conscious allocation logic, and atomic reservation is more responsive and more clinically appropriate than a system based on proximity-only allocation. It is computationally lightweight, architecturally transparent, and parameterized with interpretable weights, which can be changed to reflect local clinical protocols without re-architecting the underlying system.

6.1 Directions for Future Research

Several directions emerge from the limitations of the current study and the findings it produces.

Real road network integration. The simulation uses Euclidean distance as a proxy for travel time. Replacing the abstract grid with a real urban road network — using routing APIs or OpenStreetMap data — would produce travel-time-based scores that account for traffic patterns, road topology, and mode of transport. This is the most important extension for translating the framework toward deployment.

Blood group compatibility modelling. The current framework allocates any available unit to any request regardless of blood type. Incorporating ABO and Rh compatibility constraints would reduce the effective supply available to each request and change the allocation dynamics substantially. Compatibility-aware scoring, as studied by Jafarkhan et al. [48] and Zahiri and Pishvaei [37], is a direct extension of the composite scoring function proposed here.

Token expiry and reallocation policy. The simulation implements reservation as an immediate, permanent inventory decrement. A production system requires a token timeout policy: if a requester does not arrive within a defined window, the reserved unit must be released for reallocation. The design of this timeout — including the trade-off between lock duration, reallocation opportunity cost, and requester transit time variability — is an open problem with direct operational significance.

Sensitivity analysis of priority weights. The weights $w_{\text{high}} = 0.6$, $w_{\text{medium}} = 1.0$, $w_{\text{low}} = 1.4$ were selected to ensure full high-priority service under scarcity while preserving load balancing. A systematic sensitivity analysis across weight combinations

would characterise the trade-off surface between high-priority protection and low-priority service, and could inform evidence-based weight calibration for specific healthcare contexts.

Dynamic inventory replenishment. The current model treats bank inventory as a fixed resource that depletes monotonically during a simulation run. Real blood banks receive replenishment from collection drives, inter-facility transshipment, and regional reserves. Incorporating stochastic replenishment — as modelled by Dehghani and Abbasi [5] and Zhou et al. [6] — would test whether the framework's triage advantage persists when supply conditions change dynamically during an emergency.

Donor-side integration. The current framework addresses allocation from existing inventory. Parikh et al. [25] demonstrated that proximity-aware donor notification can reduce emergency search time on the supply side. Integrating donor recruitment logic with the allocation framework — so that a critically low bank triggers nearby donor notifications while simultaneously routing requests to better-stocked alternatives — would extend the framework into a two-sided emergency blood coordination system.

Validation against operational data. The framework has been validated through repeated simulation under varied environmental conditions. Validation against real blood bank operational records — matching allocation outcomes against actual demand patterns and inventory records — would substantially strengthen the generalisability of the findings and identify any systematic divergence between the abstract simulation environment and real operational conditions.

Scalability to regional and national networks. The current simulation uses 20 blood banks on a single grid. Regional and national blood supply networks involve hundreds of facilities across heterogeneous geographic areas with hierarchical transshipment structures. Testing the framework at scale, potentially incorporating the multi-echelon structures studied by Khiabani et al. [36], would determine whether the priority-ordering and scoring mechanisms remain computationally tractable and clinically effective at larger network sizes.

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- **Funding:** Not applicable.
- **Conflict of interest:** The authors declare that they have no conflict of interest.
- **Ethics approval and consent to participate:** Not applicable. This study involves computational simulation only and does not involve

human participants, human tissue, or animal subjects.

- **Consent for publication:** All authors have read and approved the final manuscript and consent to its publication.
- **Data availability:** The simulation framework and generated datasets are available from the corresponding author upon reasonable request.
- **Materials availability:** Not applicable.
- **Code availability:** The Python simulation code implementing the GeoAI Token allocation framework and the baseline nearest-neighbour model is available from the corresponding author upon reasonable request.
- **Author contributions:** Priyanka Asthana designed the simulation framework, implemented the GeoAI Token allocation model and baseline, conducted all experiments, generated the results, and wrote the manuscript. Dr. Laxmi Shanker Maurya supervised the research, provided conceptual guidance, and reviewed the manuscript. Harshita Kushwaha contributed to the experimental design and manuscript review. All authors read and approved the final version.

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